

yolov8_confussion_matrix

July 21, 2025

```
[1]: import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import glob
import os
```

Configuration

```
[2]: # Path configuration
VALIDATION_LABELS_PATH = 'yolo_dataset/labels/val'
PREDICTION_RESULTS_PATH = 'runs/detect/yolov8n_v8_5e_predict'
VALIDATION_RESULTS_PATH = 'runs/detect/yolov8n_v8_5e_val'

# Model configuration
CONFIDENCE_THRESHOLD = 0.5
CLASS_NAMES = ['pothole']

# Plot settings
plt.style.use('default')
sns.set_palette("husl")
```

Helper Functions

```
[3]: def load_ground_truth_labels(label_dir):
    """Load ground truth labels dari file YOLO format"""
    gt_labels = []
    label_files = glob.glob(os.path.join(label_dir, '*.txt'))

    print(f"Ditemukan {len(label_files)} file label ground truth")

    for label_file in label_files:
        with open(label_file, 'r') as f:
            for line in f.readlines():
                parts = line.strip().split()
                if len(parts) == 5:
                    class_id = int(parts[0])
                    gt_labels.append(class_id)

    print(f"Total ground truth potholes: {len(gt_labels)}")
```

```
return gt_labels
```

```
[4]: def load_predicted_labels(pred_dir, confidence_threshold=0.5):
      """Load predicted labels dari hasil inference YOLOv8"""
      pred_labels = []
      pred_files = glob.glob(os.path.join(pred_dir, 'labels', '*.txt'))

      print(f"Ditemukan {len(pred_files)} file prediksi")

      for pred_file in pred_files:
          if os.path.getsize(pred_file) > 0:
              with open(pred_file, 'r') as f:
                  for line in f.readlines():
                      parts = line.strip().split()
                      if len(parts) >= 6:
                          class_id = int(parts[0])
                          confidence = float(parts[5])
                          if confidence >= confidence_threshold:
                              pred_labels.append(class_id)

      print(f"Total prediksi dengan confidence >= {confidence_threshold}:
      ↪{len(pred_labels)}")
      return pred_labels
```

Load Data

```
[5]: print(" Loading ground truth labels...")
      gt_labels = load_ground_truth_labels(VALIDATION_LABELS_PATH)

      print("\n Loading predicted labels...")
      pred_labels = load_predicted_labels(PREDICTION_RESULTS_PATH,
      ↪CONFIDENCE_THRESHOLD)
```

Loading ground truth labels...
Ditemukan 133 file label ground truth
Total ground truth potholes: 299

Loading predicted labels...
Ditemukan 125 file prediksi
Total prediksi dengan confidence >= 0.5: 0

Data Summary

```
[ ]: print("\n" + "="*50)
      print(" DATA SUMMARY")
      print("="*50)
      print(f"Total Ground Truth Potholes: {len(gt_labels)}")
      print(f"Total Predicted Potholes: {len(pred_labels)}")
```

```

if len(gt_labels) > 0:
    print("\nGround Truth Class Distribution:")
    unique_gt, counts_gt = np.unique(gt_labels, return_counts=True)
    for cls, count in zip(unique_gt, counts_gt):
        print(f" Class {cls} (pothole): {count}")

if len(pred_labels) > 0:
    print("\nPredicted Class Distribution:")
    unique_pred, counts_pred = np.unique(pred_labels, return_counts=True)
    for cls, count in zip(unique_pred, counts_pred):
        print(f" Class {cls} (pothole): {count}")

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DATA SUMMARY
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Total Ground Truth Potholes: 299
Total Predicted Potholes: 0

```

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Ground Truth Class Distribution:
  Class 0 (pothole): 299

```

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## YOLOv8 Built-in Confusion Matrix

```

```

[ ]: def show_yolo_confusion_matrix():
    """Tampilkan confusion matrix built-in dari YOLOv8"""
    cm_path = os.path.join(VALIDATION_RESULTS_PATH, 'confusion_matrix.png')

    if os.path.exists(cm_path):
        print(" YOLOv8 Built-in Confusion Matrix:")
        img = plt.imread(cm_path)

        plt.figure(figsize=(12, 10))
        plt.imshow(img)
        plt.axis('off')
        plt.title('YOLOv8 Built-in Confusion Matrix', fontsize=16,
↪fontweight='bold')
        plt.tight_layout()
        plt.show()

        print(f" Confusion matrix loaded from: {cm_path}")
    else:
        print(f" YOLOv8 confusion matrix tidak ditemukan di: {cm_path}")
        print(" Pastikan sudah menjalankan validasi dengan mode=val")

    # List available files
    if os.path.exists(VALIDATION_RESULTS_PATH):

```

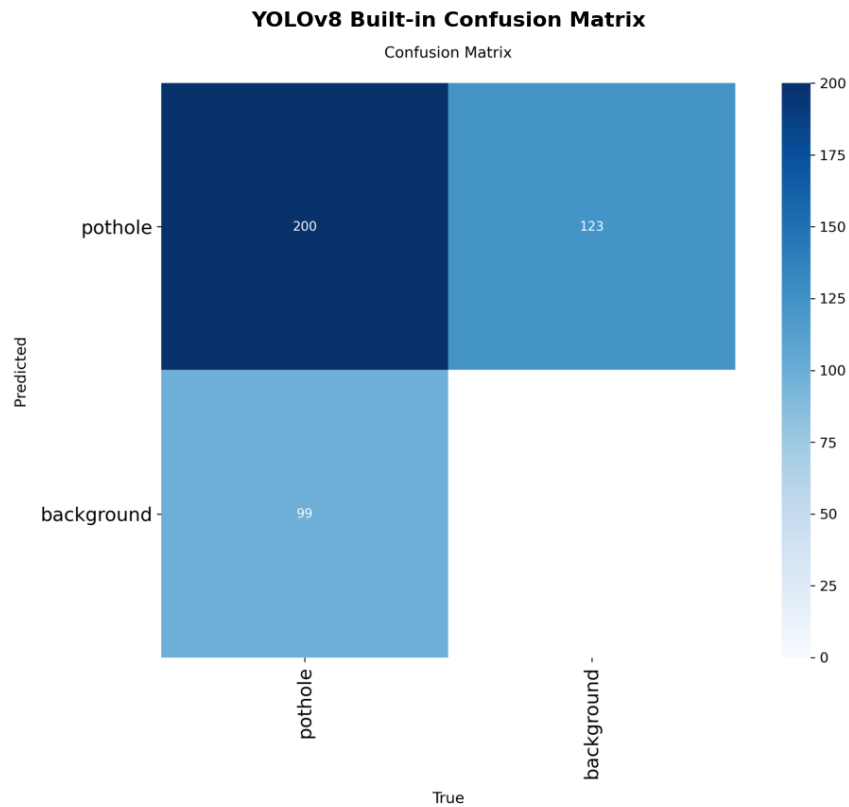
```

available_files = os.listdir(VALIDATION_RESULTS_PATH)
print(" File yang tersedia:")
for file in available_files:
    print(f" - {file}")

# Show YOLOv8's confusion matrix
show_yolo_confusion_matrix()

```

YOLOv8 Built-in Confusion Matrix:



Confusion matrix loaded from:
runs/detect/yolov8n_v8_5e_val/confusion_matrix.png